

Module 2: Random Variables

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1 Random variables and their distribution functions

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a given probability space. In some situations we may not be directly interested in the sample space Ω ; rather we may be interested in some numerical aspect of Ω .

Example 1.1. *A fair coin (heads and tails are equally likely) is tossed three times independently. Thus*

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

and

$$\mathbb{P}(\{\omega\}) = \frac{1}{8}, \omega \in \Omega.$$

Suppose that we are interested in number of heads in three tosses, that is we are interested in the function $X : \Omega \rightarrow \mathbb{R}$ defined by

$$X(\omega) = \begin{cases} 0, & \text{if } \omega = TTT \\ 1, & \text{if } \omega \in \{HTT, THT, TTH\} \\ 2, & \text{if } \omega \in \{HHT, HTH, THH\} \\ 3, & \text{if } \omega = HHH. \end{cases}$$

Clearly the values assumed by X are random with probability

$$\begin{aligned} \mathbb{P}(X = 0) = \mathbb{P}(X = 3) &= \frac{1}{8} \\ \mathbb{P}(X = 1) = \mathbb{P}(X = 2) &= \frac{3}{8}. \end{aligned}$$

Hence $\mathbb{P}(X \in \{0, 1, 2, 3\}) = 1$.

For a function $Y : \Omega \rightarrow \mathbb{R}$ and $A \subseteq \mathbb{R}$, define

$$Y^{-1}(A) = \{\omega \in \Omega : Y(\omega) \in A\}.$$

Then it is straightforward to prove the following results.

Proposition 1. *Let $A \subseteq \mathbb{R}$, $B \subseteq \mathbb{R}$ and A_α , $\alpha \in \Lambda$, where Λ is an arbitrary index set. Let $Y : \Omega \rightarrow \mathbb{R}$ be a given function. Then*

- (a) *If $A \cap B = \emptyset$, then $Y^{-1}(A) \cap Y^{-1}(B) = \emptyset$.*
- (b) *$Y^{-1}(A^c) = (Y^{-1}(A))^c$ that is $(Y^{-1}(\mathbb{R} - A) = Y^{-1}(\mathbb{R}) - Y^{-1}(A) = \Omega - Y^{-1}(A))$*
- (c) *$Y^{-1}(\cup_{\alpha \in \Lambda} A_\alpha) = \cup_{\alpha \in \Lambda} Y^{-1}(A_\alpha)$*
- (d) *$Y^{-1}(\cap_{\alpha \in \Lambda} A_\alpha) = \cap_{\alpha \in \Lambda} Y^{-1}(A_\alpha)$.*

Definition 1.1. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a given probability function. A real valued function $X : \Omega \rightarrow \mathbb{R}$ (defined on sample space Ω) is called a random variable (RV) if for all $A \subseteq \mathbb{R}$

$$X^{-1}(A) = \{\omega \in \Omega : X(\omega) \in A\} \in \mathcal{F}.$$

Thus one can define a set function $\mathbb{P}_X : \mathcal{B} \rightarrow [0, 1]$ by

$$\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in B\}), \quad B \in \mathcal{B}$$

where $\mathcal{B}$ is some class of subsets of $\mathbb{R}$. As before, for all practical purposes we will take $\mathcal{B}$ to be power set of $\mathbb{R}$. We will simply write

$$\mathbb{P}_X(B) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in B\}) = \mathbb{P}(X \in B), \quad B \in \mathcal{B}.$$

We have the following scenario

$$(\Omega, \mathcal{F}, \mathbb{P}) \xrightarrow{X} (\mathbb{R}, \mathcal{B}, \mathbb{P}_X).$$

Theorem 1.1 (Induced probability space/measure). $(\mathbb{R}, \mathcal{B}, \mathbb{P}_X)$ is a probability, that is $\mathbb{P}_X$ is a probability function defined on $\mathcal{B}$.

Proof.

$$\begin{aligned} \mathbb{P}_X(\mathbb{R}) &= \mathbb{P}(X^{-1}(\mathbb{R})) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in \mathbb{R}\}) \\ &= \mathbb{P}(\Omega) = 1. \end{aligned}$$

Also, for any $B \in \mathcal{B}$, $\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B)) \geq 0$.

Let $\{B_n\}_{n \geq 1}$ be a collection of mutually exclusive events in $\mathcal{B}$. Then

$$\begin{aligned} \mathbb{P}_X(\cup_{n=1}^{\infty} B_n) &= \mathbb{P}(X^{-1}(\cup_{n=1}^{\infty} B_n)) = \mathbb{P}(\cup_{n=1}^{\infty} X^{-1}(B_n)), \quad (\text{Proposition 1 (c)}) \\ &= \sum_{n=1}^{\infty} \mathbb{P}(X^{-1}(B_n)), \quad (\text{since } \mathbb{P} \text{ is a probability measure}) \\ &= \sum_{n=1}^{\infty} \mathbb{P}_X(B_n). \end{aligned}$$

□

Definition 1.2. The probability function $\mathbb{P}_X$ defined above is called the probability function/measure induced by RV X and $(\mathbb{R}, \mathcal{B}, \mathbb{P}_X)$ is called the probability space induced by RV X .

The induced probability measure $\mathbb{P}_X$ describes the random behavior of X .

Example 1.2. Consider the probability space defined in Example 1.1 where

$$\Omega = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$$

$$\mathbb{P}(\{\omega\}) = \frac{1}{8}, \quad \text{for all } \omega \in \Omega$$

and $X : \Omega \rightarrow \mathbb{R}$ (number of heads in three tosses) defined by

$$X(\omega) = \begin{cases} 0, & \text{if } \omega = TTT \\ 1, & \text{if } \omega \in \{HTT, THT, TTH\} \\ 2, & \text{if } \omega \in \{HHT, HTH, THH\} \\ 3, & \text{if } \omega = HHH. \end{cases}$$

Obviously, $X : \Omega \rightarrow \mathbb{R}$ is a RV with induced probability space given by $(\mathbb{R}, \mathcal{B}, \mathbb{P}_X)$, where

$$\begin{aligned}\mathbb{P}_X(\{0\}) &= \mathbb{P}(\{TTT\}) = \frac{1}{8} \\ \mathbb{P}_X(\{1\}) &= \mathbb{P}(\{HTT, THT, TTH\}) = \frac{3}{8} \\ \mathbb{P}_X(\{2\}) &= \mathbb{P}(\{HHT, HTH, THH\}) = \frac{3}{8} \\ \mathbb{P}_X(\{3\}) &= \mathbb{P}(\{HHH\}) = \frac{1}{8}\end{aligned}$$

and for any $B \in \mathcal{B}$,

$$\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(\omega \in \Omega : X(\omega) \in B) = \sum_{i \in B} \mathbb{P}_X(\{i\}).$$

Definition 1.3. Let X be a RV defined on probability space and let $(\mathbb{R}, \mathcal{B}, \mathbb{P}_X)$ denote the probability space induced by X . Define the function $F_X : \mathbb{R} \rightarrow \mathbb{R}$ by

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{P}_X((-\infty, x]), \quad x \in \mathbb{R}.$$

The function F_X is called the cumulative distribution function (CDF) or simply the distribution function (DF) of RV X .

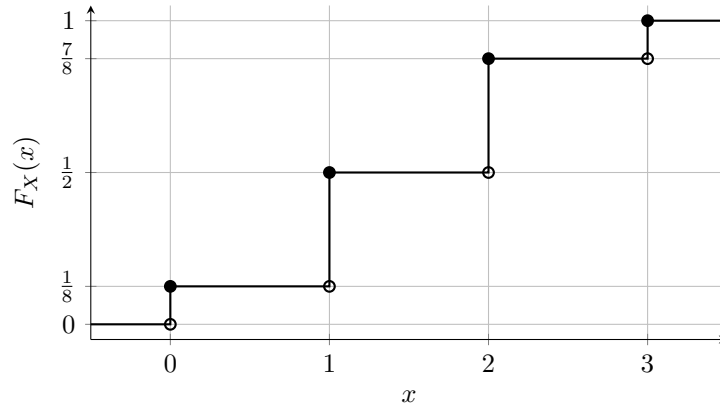
Whenever there is no ambiguity we will drop subscript X in F_X to represent DF of a RV by F . It can be shown (in advance courses) that the CDF $F_X(\cdot)$ of a RV X determine the induced probability measure $\mathbb{P}_X(\cdot)$ uniquely. Thus to study the random behavior of RV X it suffices to study its DF F .

Example 1.3. Let us revisit Examples 1.1 and 1.2, so that

$$\begin{aligned}\mathbb{P}(X = 0) = \mathbb{P}_X(\{0\}) &= \mathbb{P}(\{TTT\}) = \frac{1}{8} \\ \mathbb{P}(X = 1) = \mathbb{P}_X(\{1\}) &= \mathbb{P}(\{HTT, THT, TTH\}) = \frac{3}{8} \\ \mathbb{P}(X = 2) = \mathbb{P}_X(\{2\}) &= \mathbb{P}(\{HHT, HTH, THH\}) = \frac{3}{8} \\ \mathbb{P}(X = 3) = \mathbb{P}_X(\{3\}) &= \mathbb{P}(\{HHH\}) = \frac{1}{8}\end{aligned}$$

Then the DF of X is

$$F_X(x) = \mathbb{P}(X \leq x) = \begin{cases} 0, & \text{if } x < 0 \\ \frac{1}{8}, & \text{if } 0 \leq x < 1 \\ \frac{1}{8} + \frac{3}{8} = \frac{1}{2}, & \text{if } 1 \leq x < 2 \\ \frac{1}{8} + \frac{3}{8} + \frac{3}{8} = \frac{7}{8}, & \text{if } 2 \leq x < 3 \\ \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1, & \text{if } x \geq 3. \end{cases}$$



Theorem 1.2. Let $F(\cdot)$ be the CDF of a RV X defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and let $(\mathbb{R}, \mathcal{B}, \mathbb{P}_X)$ be the probability space induced by X . Then

- (a) F is non decreasing
- (b) F is right continuous
- (c) $F(-\infty) = \lim_{n \uparrow \infty} F(-n) = 0$
- (d) $F(\infty) = \lim_{n \uparrow \infty} F(n) = 1$.

Conversely any function $G(\cdot)$ satisfying properties (a)–(d) is a DF of some RV Y defined on a probability space $(\Omega^*, \mathcal{F}^*, \mathbb{P}^*)$.

Proof. (a) Let $-\infty < x < y < \infty$. Then $(-\infty, x] \subseteq (-\infty, y]$. It implies that $\mathbb{P}_X((-\infty, x]) \leq \mathbb{P}_X((-\infty, y]) \implies F(x) \leq F(y)$.

(b) Since F is monotone and bounded below (by 0) $\lim_{h \downarrow 0} F(x+h) = F(x+)$ exists for all $x \in \mathbb{R}$. Therefore,

$$F(x+) = \lim_{h \downarrow 0} F(x+h) = \lim_{n \rightarrow \infty} F(x + \frac{1}{n}) = \lim_{n \rightarrow \infty} \mathbb{P}_X((-\infty, x + \frac{1}{n}]).$$

Let $A_n = (-\infty, x + \frac{1}{n}]$, $n = 1, 2, \dots$. Then $A_n \downarrow$ and $\cap_{n=1}^{\infty} (-\infty, x + \frac{1}{n}] = (-\infty, x]$. Thus

$$F(x+) = \lim_{n \rightarrow \infty} \mathbb{P}_X((-\infty, x + \frac{1}{n}]) = \mathbb{P}_X(\cap_{n=1}^{\infty} (-\infty, x + \frac{1}{n}]) = \mathbb{P}_X((-\infty, x]) = F(x).$$

(c)

$$\begin{aligned} F(-\infty) &= \lim_{n \rightarrow \infty} F_X(-n) = \lim_{n \rightarrow \infty} \mathbb{P}_X((-\infty, -n]) = \mathbb{P}_X(\cap_{n=1}^{\infty} (-\infty, -n]), \text{ (since } (-\infty, -n] \downarrow) \\ &= \mathbb{P}_X(\emptyset), \text{ (since } \cap_{n=1}^{\infty} (-\infty, -n] = \emptyset) \\ &= 0. \end{aligned}$$

(d)

$$\begin{aligned} F(\infty) &= \lim_{n \rightarrow \infty} F_X(n) = \lim_{n \rightarrow \infty} \mathbb{P}_X((-\infty, n]) = \mathbb{P}_X(\cup_{n=1}^{\infty} (-\infty, n]), \text{ (since } (-\infty, n] \uparrow) \\ &= \mathbb{P}_X(\Omega), \text{ (since } \cup_{n=1}^{\infty} (-\infty, n] = \Omega) \\ &= 1. \end{aligned}$$

□

Remark 1.1. (a) Since any distribution function is monotone and bounded above (by 1) $\lim_{h \downarrow 0} F(x-h) = F(x-)$ exists for all $x \in \mathbb{R}$. Moreover,

$$\begin{aligned} F(x-) &= \lim_{h \downarrow 0} F(x-h) = \lim_{n \rightarrow \infty} F(x - \frac{1}{n}) = \lim_{n \rightarrow \infty} \mathbb{P}_X((x - \frac{1}{n}, \infty)) \\ &= \mathbb{P}_X(\cup_{n=1}^{\infty} (x - \frac{1}{n}, \infty)), \text{ } ((x - \frac{1}{n}, \infty) \uparrow) \\ &= \mathbb{P}((-\infty, x)) = \mathbb{P}_X(X < x). \end{aligned}$$

(b) Since, for any $x \in \mathbb{R}$, $F(x+)$ and $F(x-)$ exist, F has only jump discontinuities ($F(x) = F(x+) > F(x-)$). Moreover since any DF is monotone it has at most countable number of discontinuities.

(c) A DF F is continuous at $a \in \mathbb{R}$ if and only if

$$F(a) = F(a-)$$

(d) For any $a \in \mathbb{R}$,

$$\mathbb{P}(x = a) = \mathbb{P}(X \leq x) - \mathbb{P}(X < x) = F(a) - F(a-).$$

Thus the DF F is continuous at $a \in \mathbb{R}$ if and only if

$$\mathbb{P}(X = a) = F(a) - F(a-) = 0.$$

(c) We have for $-\infty < a < b < \infty$

$$\begin{aligned}\mathbb{P}(a < X \leq b) &= \mathbb{P}(X \leq b) - \mathbb{P}(X \leq a) = F(b) - F(a) \\ \mathbb{P}(a < X < b) &= \mathbb{P}(X < b) - \mathbb{P}(X \leq a) = F(b-) - F(a) \\ \mathbb{P}(a \leq X < b) &= \mathbb{P}(X < b) - \mathbb{P}(X < a) = F(b-) - F(a-) \\ \mathbb{P}(a \leq X \leq b) &= \mathbb{P}(X \leq b) - \mathbb{P}(X < a) = F(b) - F(a-) \\ \mathbb{P}(X < a) &= F(a-) \\ \mathbb{P}(X \geq a) &= 1 - F(a-).\end{aligned}$$

It can be shown that (in advanced courses) that the DF F determine the induced probability function $\mathbb{P}_X(\cdot)$ uniquely. Thus it suffices to study the DF of a RV.

Example 1.4. Consider the function $F: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F_X(x) = \mathbb{P}(X \leq x) \begin{cases} 0, & \text{if } x < 0 \\ \frac{x}{3}, & \text{if } 0 \leq x < 1 \\ \frac{1}{2}, & \text{if } 1 \leq x < 2 \\ \frac{2}{3}, & \text{if } 2 \leq x < 3 \\ 1, & \text{if } x \geq 3. \end{cases}$$

(a) Show that F is a DF of some random variable Y ;

(b) Find $\mathbb{P}(X = a)$ for various values of $a \in \mathbb{R}$;

(c) Find $\mathbb{P}(X < 3), \mathbb{P}(X \geq 1/2), \mathbb{P}(2 < X \leq 4), \mathbb{P}(1 \leq X < 2), \mathbb{P}(2 \leq X \leq 3)$ and $\mathbb{P}(1/2 < x < 3)$.

Solution: (a) Clearly F is non-decreasing in $(-\infty, 0), (0, 1), (2, 3)$ and $(3, \infty)$. Moreover,

$$\begin{aligned}F(0) - F(0-) &= 0 \geq 0 \\ F(1) - F(1-) &= \frac{1}{2} - \frac{1}{3} > 0 \\ F(2) - F(2-) &= \frac{2}{3} - \frac{1}{2} > 0 \\ F(3) - F(3-) &= 1 - \frac{2}{3} > 0\end{aligned}$$

It follows that F is non-decreasing.

Clearly F is continuous (and hence right continuous) on $(-\infty, 0), (0, 1), (2, 3)$ and $(3, \infty)$. Moreover,

$$\begin{aligned}F(0+) - F(0) &= 0 - 0 = 0 \\ F(1+) - F(1) &= \frac{1}{2} - \frac{1}{2} = 0 \\ F(2+) - F(2) &= \frac{2}{3} - \frac{2}{3} = 0 \\ F(3+) - F(3) &= 1 - 1 = 0\end{aligned}$$

$\implies F$ is right continuous on $\mathbb{R}$.

(b) The set of discontinuity points of F is $D = \{1, 2, 3\}$. Thus

$$\begin{aligned}\mathbb{P}(X = a) = F(a) - F(a-) &= 0, \text{ for all } a \neq 1, 2, 3 \\ \mathbb{P}(X = 1) = F(1) - F(1-) &= \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \\ \mathbb{P}(X = 2) = F(2) - F(2-) &= \frac{2}{3} - \frac{1}{2} = \frac{1}{6} \\ \mathbb{P}(X = 3) = F(3) - F(3-) &= 1 - \frac{2}{3} = \frac{1}{3}\end{aligned}$$

(c)

$$\begin{aligned}\mathbb{P}(X < 3) = F(3-) &= \frac{2}{3} \\ \mathbb{P}(X \geq 1/2) = 1 - F(\frac{1}{2}-) &= 1 - \frac{1}{6} = \frac{5}{6} \\ \mathbb{P}(2 < X \leq 4) = F(4) - F(2) &= 1 - \frac{2}{3} = \frac{1}{3} \\ \mathbb{P}(1 \leq X < 2) = F(2-) - F(1-) &= \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \\ \mathbb{P}(2 \leq X \leq 3) = F(3) - F(2-) &= 1 - \frac{1}{2} = \frac{1}{2} \\ \mathbb{P}(\frac{1}{2} < X < 3) = F(3-) - F(\frac{1}{2}) &= \frac{2}{3} - \frac{1}{6} = \frac{1}{2}.\end{aligned}$$

2 Discrete random variables

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \rightarrow \mathbb{R}$ be a RV with induced probability space $(\mathbb{R}, \mathcal{B}, \mathbb{P}_X)$ and DF F .

Definition 2.1. The RV X is said to be a discrete RV if there exists a countable set S (finite or countably infinite) such that

$$\mathbb{P}(X = x) = F(x) - F(x-) > 0, \text{ for all } x \in S$$

and

$$\mathbb{P}(X \in S) = 1.$$

The set S is called the support of RV X .

Remark 2.1. (a) If S is the support of a discrete RV X then clearly

$$S = \{x \in \mathbb{R} : F(x) - F(x-) > 0\}$$

is the set of discontinuity points of F since $\mathbb{P}(X \in S) = 1$ and $F(x) - F(x-) > 0$ for all $x \in S$.

(b) Note that if x is a discontinuity point of DF F then

$$F(x) - F(x-) = \text{size of jump of } F \text{ at } x$$

$$\text{since } \mathbb{P}(X \in S) = \sum_{x \in S} \mathbb{P}(X = x) = \sum_{x \in S} [F(x) - F(x-)] = 1.$$

Example 2.1. In Example 1.4 the set of discontinuity points of F is $D = \{1, 2, 3\}$ and

$$\sum_{x \in D} [F(x) - F(x-)] = \frac{1}{6} + \frac{1}{6} + \frac{1}{3} = \frac{2}{3} < 1.$$

It implies that Y is not a discrete RV.

Example 2.2. Consider the DF (Example 1.3)

$$G(x) = \mathbb{P}(X \leq x) = \begin{cases} 0, & \text{if } x < 0 \\ \frac{1}{8}, & \text{if } 0 \leq x < 1 \\ \frac{1}{2}, & \text{if } 1 \leq x < 2 \\ \frac{7}{8}, & \text{if } 2 \leq x < 3 \\ 1, & \text{if } x > 3. \end{cases}$$

The set of discontinuity points of G is $D = \{0, 1, 2, 3\}$ with

$$\sum_{x \in D} [F(x) - F(x-)] = \frac{1}{8} + \left(\frac{1}{2} - \frac{1}{8}\right) + \left(\frac{7}{8} - \frac{1}{2}\right) + \left(1 - \frac{7}{8}\right) = 1.$$

$\implies X$ is a discrete RV with support $S = D = \{0, 1, 2, 3\}$.

Definition 2.2. Let X be a RV with CDF F_X and support S_X . Define the function $f_X : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f_X = \begin{cases} \mathbb{P}(X = x) = F_X(x) - F_X(x-), & \text{if } x \in S \\ 0, & \text{Otherwise.} \end{cases}$$

The function f_x is called the probability mass function (pmf) of RV X .

Whenever there is no ambiguity we will drop the subscript X in F_X , S_X and f_X to represent the DF of X by F , the support of X by S and the PMF of X by f .

Remark 2.2. (a) Let X be a discrete RV with PMF f and DF F . Let S be the support of X . Then for any $A \subseteq \mathbb{R}$,

$$\mathbb{P}(X \in A \cap S) = \sum_{x \in A \cap S} f(x).$$

Moreover,

$$F(x) = \sum_{y \in S \cap (-\infty, x]} f(y).$$

Further, for any $x \in S$, $f(x) = F(x) - F(x-)$.

(b) Clearly a DF determines the PMF uniquely and vice-versa. Thus it suffices to study the PMF of a discrete RV.

(c) Let X be a discrete RV with PMF f and support S . Then $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies

- (i) $f(x) \geq 0$, for all $x \in S$
- (ii) $\sum_{x \in S} f(x) = 1$.

Conversely, suppose that $g : \mathbb{R} \rightarrow \mathbb{R}$ is a function such that, for some countable set T ,

- (i) $g(x) \geq 0$, for all $x \in T$
- (ii) $\sum_{x \in T} g(x) = 1$.

Then $g(\cdot)$ is the PMF of some discrete RV having support T .

Example 2.3. Let Y be a RV having DF

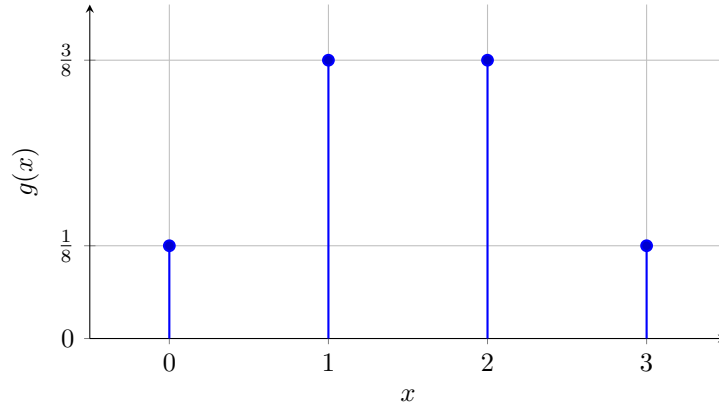
$$G(x) = \mathbb{P}(X \leq x) = \begin{cases} 0, & \text{if } x < 0 \\ \frac{1}{8}, & \text{if } 0 \leq x < 1 \\ \frac{1}{2}, & \text{if } 1 \leq x < 2 \\ \frac{7}{8}, & \text{if } 2 \leq x < 3 \\ 1, & \text{if } x > 3. \end{cases}$$

We have seen in Example 2.2 that Y is a discrete RV with support $S = \{0, 1, 2, 3\}$. The PMF of Y is $g : \mathbb{R} \rightarrow \mathbb{R}$ where

$$\begin{aligned} g(0) &= G(0) - G(0-) = \frac{1}{8} - 0 = \frac{1}{8} \\ g(1) &= G(1) - G(1-) = \frac{1}{2} - \frac{1}{8} = \frac{3}{8} \\ g(2) &= G(2) - G(2-) = \frac{7}{8} - \frac{1}{2} = \frac{3}{8} \\ g(3) &= G(3) - G(3-) = 1 - \frac{7}{8} = \frac{1}{8}. \end{aligned}$$

Thus the PMF of Y is DF

$$g(x) = \begin{cases} \frac{1}{8}, & \text{if } x = 0, 3 \\ \frac{3}{8}, & \text{if } x = 1, 2 \\ 0, & \text{Otherwise.} \end{cases}$$



Example 2.4. A fair die (all outcomes are equally likely) is tossed repeatedly and independently until a 6 is observed. Then X is a discrete RV with support $S = \{1, 2, \dots, 6\}$. The PMF

$$f(x) = \mathbb{P}(X = x) = \begin{cases} \left(\frac{5}{6}\right)^{x-1} \frac{1}{6}, & \text{if } x = 1, 2, \dots, 6 \\ 0, & \text{Otherwise.} \end{cases}$$

and DF

$$F(x) = \mathbb{P}(X \leq x) = \begin{cases} 0, & \text{if } x < 1 \\ \frac{1}{6}, & \text{if } 1 \leq x < 2 \\ \frac{11}{36}, & \text{if } 2 \leq x < 3 \\ \sum_{j=1}^i \left(\frac{5}{6}\right)^{j-1} \frac{1}{6} = 1 - \left(\frac{5}{6}\right)^i, & \text{if } i \leq x < i+1 \\ \vdots & \end{cases}$$

3 Continuous random variable

Let X be a RV with DF F .

Definition 3.1. The RV X is said to be a continuous RV if there exists a non-negative integral function $f : \mathbb{R} \rightarrow [0, \infty)$ such that, for any $x \in \mathbb{R}$,

$$F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f(t) dt.$$

The function $f(\cdot)$ is called the probability density function (PDF) of X . The support of the continuous RV X is the set $S = \{x \in \mathbb{R} : F(x+h) - F(x-h) > 0, \forall h > 0\}$.